MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics | **PERIODIC ASSESSMENT – 09** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 8: Introduction to Trigonometry** | **MM: 30**

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper units where applicable.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Objective Type Questions

MATHS MANTHAN | (6 × 1 = 6 Marks)

Q1. The value of $\dfrac{1}{1+\sin\theta}+\dfrac{1}{1-\sin\theta}$ is equal to: [1]

(A) $2\sec^2\theta$ (B) $2\cos^2\theta$

(C) $2\sin^2\theta$ (D) $\sec^2\theta$

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) $2\sec^2\theta$

Justification: $\dfrac{(1-\sin\theta)+(1+\sin\theta)}{1-\sin^2\theta}=\dfrac{2}{\cos^2\theta}=2\sec^2\theta.$

Q2. If $\csc\theta-\cot\theta=\dfrac{1}{3}$, then the value of $\csc\theta+\cot\theta$ is: [1]

(A) $\dfrac{1}{3}$ (B) 3 (C) $\dfrac{2}{3}$ (D) 1

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) 3

Justification: Using $\csc^2\theta-\cot^2\theta=1$, $\csc\theta+\cot\theta=\dfrac{1}{\csc\theta-\cot\theta}=\dfrac{1}{1/3}=3.$

Q3. The expression $(\sin^4\theta - \cos^4\theta + 1)\csc^2\theta$ simplifies to: [1]

(A) 1 (B) 2 (C) 0 (D) -1

► STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (B) 2

Justification: $\sin^4\theta - \cos^4\theta = (\sin^2\theta - \cos^2\theta)(\sin^2\theta + \cos^2\theta) = 2\sin^2\theta - 1$. Adding 1 gives $2\sin^2\theta$, and $(2\sin^2\theta)\csc^2\theta = 2$.

Q4. If $k + 1 = \sec^2\theta(1 + \sin\theta)(1 - \sin\theta)$, then the value of k is: [1]

(A) 0 (B) 1 (C) 2 (D) -1

► STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (A) 0

Justification: RHS $= \sec^2\theta(1 - \sin^2\theta) = \sec^2\theta \cdot \cos^2\theta = 1$. Thus $k + 1 = 1 \implies k = 0$.

Q5. The value of $\sqrt{\dfrac{1 + \cos\theta}{1 - \cos\theta}}$ for an acute angle θ is equal to: [1]

(A) $\csc\theta + \cot\theta$ (B) $\sec\theta + \tan\theta$

(C) $\csc\theta - \cot\theta$ (D) $\sec\theta - \tan\theta$

► STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (A) $\csc\theta + \cot\theta$

Justification: $\sqrt{\dfrac{(1 + \cos\theta)^2}{1 - \cos^2\theta}} = \dfrac{1 + \cos\theta}{\sin\theta} = \csc\theta + \cot\theta$ (for acute θ).

Q6. **Assertion (A):** In a right-angled triangle ABC with $\angle B = 90°$, $\sin A = \cos C$.
Reason (R): $\sin A = \dfrac{BC}{AC}$ and $\cos C = \dfrac{BC}{AC}$, so both are identical side ratios. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

► STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (A)

Justification: Opposite side to $\angle A$ is BC, which is also the adjacent side to $\angle C$; thus both ratios equal $\dfrac{BC}{AC}$. Both are true and (R) explains (A).

SECTION B: Very Short Answer Type — MATHS MANTHAN | (2 × 2 = 4 Marks)

Q7. Prove the identity: $(\sin\theta + \cos\theta)^2 + (\sin\theta - \cos\theta)^2 = 2$. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Expand Both Squares [1 Mark]

$$\text{LHS} = (\sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta) + (\sin^2\theta + \cos^2\theta - 2\sin\theta\cos\theta)$$

Step 2: Combine Using $\sin^2\theta + \cos^2\theta = 1$ [1 Mark]

$$\text{LHS} = 2(\sin^2\theta + \cos^2\theta) = 2(1) = \boxed{\mathbf{2} = \text{RHS} \quad \text{[Hence Proved]}}$$

Q8. Express $\sin A$, $\sec A$, and $\tan A$ in terms of $\cot A$. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Express $\sin A$ **via** $\csc^2 A = 1 + \cot^2 A$ [1 Mark]

For acute A, $\sin A = \dfrac{1}{\csc A} = \dfrac{1}{\sqrt{1+\cot^2 A}}$. **Step 2: Express** $\tan A$ **and** $\sec A$ [1 Mark]

$$\tan A = \frac{1}{\cot A}, \qquad \sec A = \sqrt{1+\tan^2 A} = \sqrt{1+\frac{1}{\cot^2 A}} = \frac{\sqrt{\cot^2 A + 1}}{\cot A}$$

$$\boxed{\sin A = \frac{1}{\sqrt{\mathbf{1} + \cot^{\mathbf{2}} \mathbf{A}}}, \quad \sec A = \frac{\sqrt{\mathbf{1} + \cot^{\mathbf{2}} \mathbf{A}}}{\cot \mathbf{A}}, \quad \tan A = \frac{\mathbf{1}}{\cot \mathbf{A}}}$$

SECTION C: Short Answer Type — MATHS MANTHAN | (2 × 3 = 6 Marks)

Q9. Prove that: $(1 + \tan A \tan B)^2 + (\tan A - \tan B)^2 = \sec^2 A \sec^2 B$. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Expand Both Squares on LHS [1.5 Marks]

$$\text{LHS} = (1 + 2\tan A \tan B + \tan^2 A \tan^2 B) + (\tan^2 A - 2\tan A \tan B + \tan^2 B)$$

$$\text{LHS} = 1 + \tan^2 A + \tan^2 B + \tan^2 A \tan^2 B = (1 + \tan^2 A) + \tan^2 B(1 + \tan^2 A)$$

Step 2: Factorize & Use $1 + \tan^2\theta = \sec^2\theta$ [1.5 Marks]

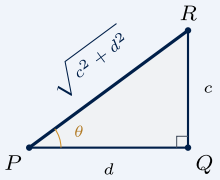
$$\text{LHS} = (1 + \tan^2 A)(1 + \tan^2 B) = \boxed{\sec^{\mathbf{2}} \mathbf{A} \sec^{\mathbf{2}} \mathbf{B} = \text{RHS} \quad \text{[Hence Proved]}}$$

Q10. If $\sin\theta = \dfrac{c}{\sqrt{c^2+d^2}}$ $(d>0)$, find the values of $\cos\theta$ and $\tan\theta$. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Step 1: Find Adjacent Side via Right Triangle [1.5 Marks]

Let $\theta = \angle P$ in right $\triangle PQR$ ($\angle Q = 90°$) with Opp $QR = c$ and Hyp $PR = \sqrt{c^2+d^2}$ ($c, d > 0$ for acute θ). By Pythagoras' theorem:



$$PQ = \sqrt{PR^2 - QR^2} = \sqrt{(c^2+d^2)-c^2} = \sqrt{d^2} = d \quad (\because d>0)$$

Step 2: Evaluate $\cos\theta$ and $\tan\theta$ [1.5 Marks]

$$\cos\theta = \frac{PQ}{PR} = \frac{d}{\sqrt{c^2+d^2}}, \qquad \tan\theta = \frac{QR}{PQ} = \frac{c}{d}$$

$$\boxed{\cos\theta = \frac{\mathbf{d}}{\sqrt{\mathbf{c^2+d^2}}}, \qquad \tan\theta = \frac{\mathbf{c}}{\mathbf{d}}}$$

SECTION D: Long Answer / HOTS

MATHS MANTHAN | **(2 × 5 = 10 Marks)**

Q11. If $\sec\theta = x + \dfrac{1}{4x}$, prove that $\sec\theta + \tan\theta = 2x$ or $\dfrac{1}{2x}$. **[5]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

Step 1: Evaluate $\tan^2\theta = \sec^2\theta - 1$ [2 Marks]

$$\tan^2\theta = \left(x+\frac{1}{4x}\right)^2 - 1 = x^2 + 2(x)\left(\frac{1}{4x}\right) + \frac{1}{16x^2} - 1 = x^2 - \frac{1}{2} + \frac{1}{16x^2} = \left(x-\frac{1}{4x}\right)^2$$

Step 2: Take Square Roots [1.5 Marks]

$$\implies \tan\theta = \pm\left(x - \frac{1}{4x}\right)$$

Step 3: Consider Both Cases for $\sec\theta + \tan\theta$ [1.5 Marks]

$$\text{Case I } (+): \quad \sec\theta + \tan\theta = \left(x+\frac{1}{4x}\right) + \left(x-\frac{1}{4x}\right) = \mathbf{2x}$$

$$\text{Case II } (-): \quad \sec\theta + \tan\theta = \left(x+\frac{1}{4x}\right) - \left(x-\frac{1}{4x}\right) = \frac{2}{4x} = \mathbf{\frac{1}{2x}}$$

$$\boxed{\therefore \sec\theta + \tan\theta = \mathbf{2x} \text{ or } \mathbf{\frac{1}{2x}} \quad \text{[Hence Proved]}}$$

Q12. If $\sin\theta + \cos\theta = p$ and $\sec\theta + \csc\theta = q$, prove that $q(p^2 - 1) = 2p$. [5]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

Step 1: Compute $p^2 - 1$ [2 Marks]

$$p^2 = (\sin\theta + \cos\theta)^2 = \sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta = 1 + 2\sin\theta\cos\theta \implies p^2 - 1 = 2\sin\theta\cos\theta$$

Step 2: Simplify q in Terms of p [1.5 Marks]

$$q = \frac{1}{\cos\theta} + \frac{1}{\sin\theta} = \frac{\sin\theta + \cos\theta}{\sin\theta\cos\theta} = \frac{p}{\sin\theta\cos\theta}$$

Step 3: Evaluate $q(p^2 - 1)$ [1.5 Marks]

$$q(p^2 - 1) = \left(\frac{p}{\sin\theta\cos\theta}\right)(2\sin\theta\cos\theta) = \boxed{\mathbf{2p} \quad \text{[Hence Proved]}}$$

SECTION E: Case-Based / Applied Situation — MATHS MANTHAN | (1 × 4 = 4 Marks)

Q13. Pedestrian Foot Overbridge Staircase Design

An urban transit planner designs a straight public staircase for a foot overbridge. The total vertical height to be climbed is 6 m. Each step has a riser (height) of 15 cm and a tread (depth) of 20 cm, establishing an inclination angle θ for the stringer beam such that $\tan\theta = \frac{15}{20} = \frac{3}{4}$. **Based on the above geometric model, answer the following questions:**

(i) Determine the values of $\sin\theta$ and $\cos\theta$ for the staircase stringer incline. [1]

(ii) Calculate the total horizontal run D of the staircase. [1]

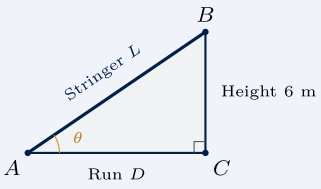
(iii) Calculate the exact length L of the diagonal handrail beam spanning from the bottom landing to the bridge platform. [2]

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]

Sub-part (i): Values of $\sin\theta$ and $\cos\theta$ [1 Mark]

For $\tan\theta = \frac{3}{4}$ (a 3–4–5 right triangle), Hypotenuse $= \sqrt{3^2 + 4^2} = 5$:

$$\boxed{\sin\theta = \frac{\mathbf{3}}{\mathbf{5}}, \qquad \cos\theta = \frac{\mathbf{4}}{\mathbf{5}}}$$



Sub-part (ii): Total Horizontal Run D [1 Mark]

$$\tan\theta = \frac{\text{Height}}{D} \implies \frac{3}{4} = \frac{6}{D} \implies \boxed{D = \mathbf{8}\text{ m}}$$

Sub-part (iii): Length L of Handrail Stringer [2 Marks]

$$L = \sqrt{D^2 + \text{Height}^2} = \sqrt{8^2 + 6^2} = \sqrt{64 + 36}$$

$$= \sqrt{100} \implies \boxed{L = \mathbf{10}\text{ m}}$$

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